entergy

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10 CFR 50.73

GNRO2024-00020

May 28, 2024

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2024-002-00, Automatic Actuation of Reactor Protection System

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report (LER) 2024-002-00, Automatic Actuation of Reactor Protection System. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of any of the systems listed in paragraph (a)(2)(iv)(B).

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact me at 802-380-5124.

Sincerely,

JAH/saw

Attachments: Licensee Event Report 2024-002-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Attachment
Licensee Event Report 2024-002-00

NRC FORM 366 (04-02-2024)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Grand Gulf Nuclear Station	■ 050 / ☐ 052	416	1 OF 2

4. Title
Automatic Actuation of Reactor Protection System

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
03	26	2024	2024	- 002 -	00	05	28	2024	N/A	☐ 050	
									N/A	☐ 052	

9. Operating Mode	10. Power Level
4	0

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☑ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Jeff Hardy, Regulatory Assurance Manager	802-380-5124

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On March 26, 2024 at 1115 CDT, Grand Gulf Nuclear Station experienced an automatic actuation of the reactor protection system (RPS) due to high reactor coolant system (RCS) pressure while performing in-service leak testing. The plant was in Mode 4 at zero percent power with all control rods inserted at the time of the RPS actuation.

The RPS actuation occurred due to a valid high RCS pressure signal and all plant systems responded as designed. The high pressure condition occurred due to inadvertent isolation of the blowdown discharge flow path.

There were no safety consequences impacting the plant or public as a result of this event. The reactor was shutdown and subcritical at the time of the actuation.

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Grand Gulf Nuclear Station	■ 050 / ☐ 052	416	2024	002	00

NARRATIVE

PLANT CONDITIONS

At the time of the event, Grand Gulf Nuclear Station (GGNS) was in Mode 4 (Cold Shutdown) with all control rods inserted at 0% reactor power. The station was performing in-service leak testing with reactor coolant system (RCS) pressure at approximately 1025 psig.

DESCRIPTION OF EVENT

During the performance of in-service leak testing, RCS pressure was being maintained with control rod drive (CRD) flow and reactor water cleanup (RWCU) blowdown to the main condenser. The Control Room was transferring RWCU blowdown to the radwaste system when RCS pressure began to rise. Reactor operators took action to reduce CRD flow, but RCS pressure did not reduce rapidly. The reactor protection system actuated as designed on high RCS pressure.

Operators responded per the off-normal event procedure and the RPS was reset.

REPORTABILITY

The automatic RPS actuation was initially reported to the NRC as a 4-hour report per 10 CFR 50.72(b)(2)(iv)(B) on March 26, 2024. This report is made in accordance with 10 CFR 50.73(a)(2)(iv)(A) for any event or condition that resulted in manual or automatic actuation of any system listed in paragraph 10 CFR 50.73(a)(2)(iv)(B).

CAUSE

The cause of the event was RWCU blowdown discharge flow path was isolated with the RCS at high pressure. Operators believed that the RWCU blowdown flow path to the main condenser could be isolated before aligning a discharge flow path to the radwaste system.

CORRECTIVE ACTIONS

Completed actions:

• The RWCU discharge flow path was re-established.
• The RPS was reset.

Planned action:

• Revise integrated operating instruction procedure 03-1-01-6 "Reactor Vessel In-Service Leak Test" to add specific guidance for swapping RWCU discharge flow paths.

SAFETY SIGNIFICANCE

There were no safety consequences, real or potential, resulting from the event. The reactor was sub-critical with all rods inserted at the time of the event and all systems responded as designed.

PREVIOUS SIMILAR EVENTS

None